Ripple

A single calibrated, transparent NLP risk engine
powering a fast index rebalancer and a deep stress tester

<Your Full Name> | <Your College>

S&P Global & Crisil Campus Hackathon 2026 - Phase III Case Study

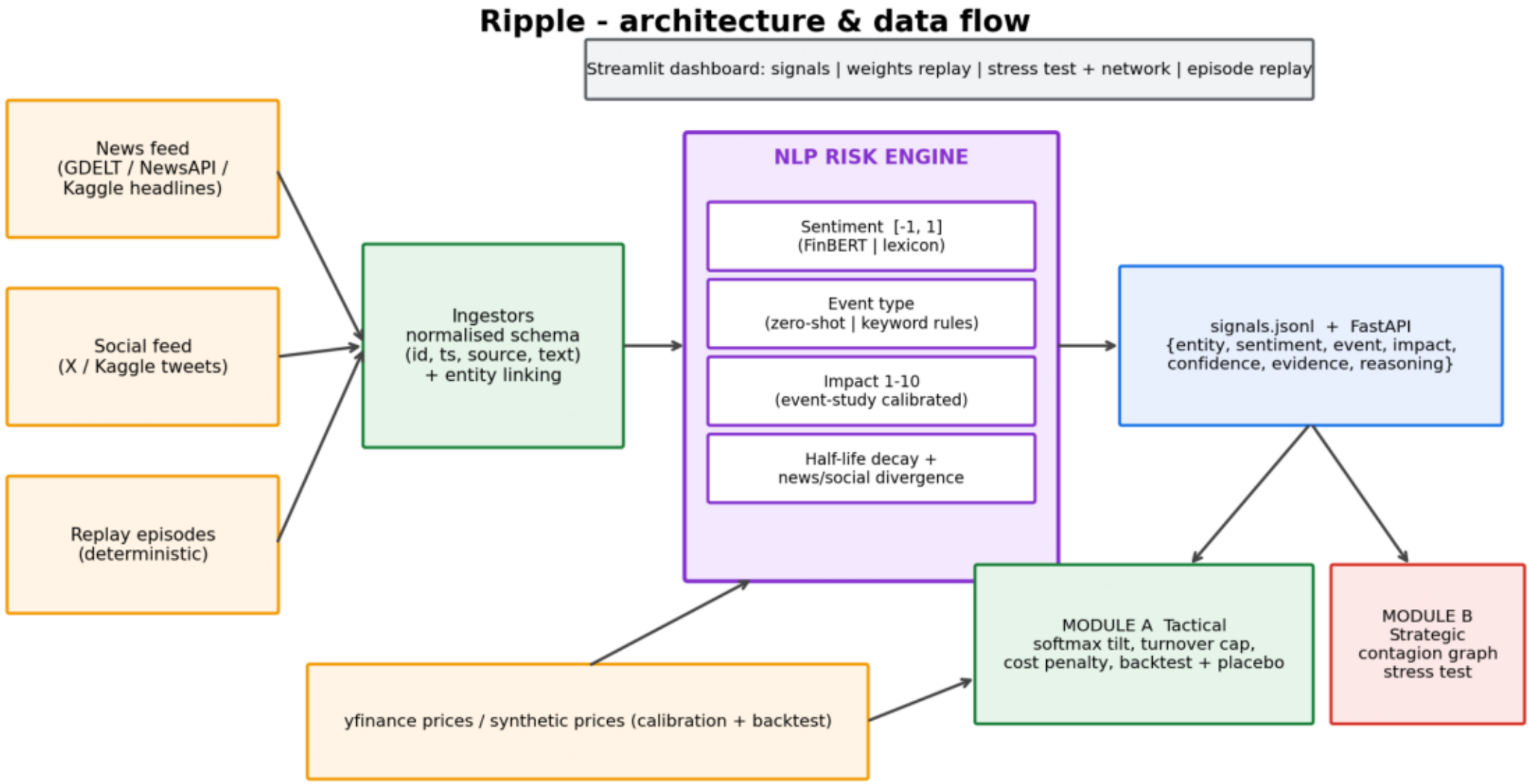
Problem & approach

The problem: news and social posts move prices faster than risk systems can digest them.

Our approach: one engine converts text into structured, auditable signals used over two horizons:
- Module A (short-term, daily): index weights tilted by sentiment
- Module B (long-term, event-driven): stress test on a wholesale-banking book

What sets it apart:
1. Impact is LEARNED from an event study rather than assumed
2. Sentiment fades per event type (half-lives), preventing erratic weights
3. A news-versus-social gap flags rumours and early warnings
4. Each signal includes evidence, confidence and reasoning
5. A contagion graph and deterministic replay mode

System design



Implementation highlights

Engine: entity linking -> sentiment [-1,1] -> event category -> impact 1-10 -> decay and divergence
  Back-ends: offline financial lexicon (default, deterministic) or FinBERT + BART-MNLI zero-shot
    (--model transformer)
  Delivery: signals.jsonl plus a FastAPI service (/signals, /analyze, /stress)

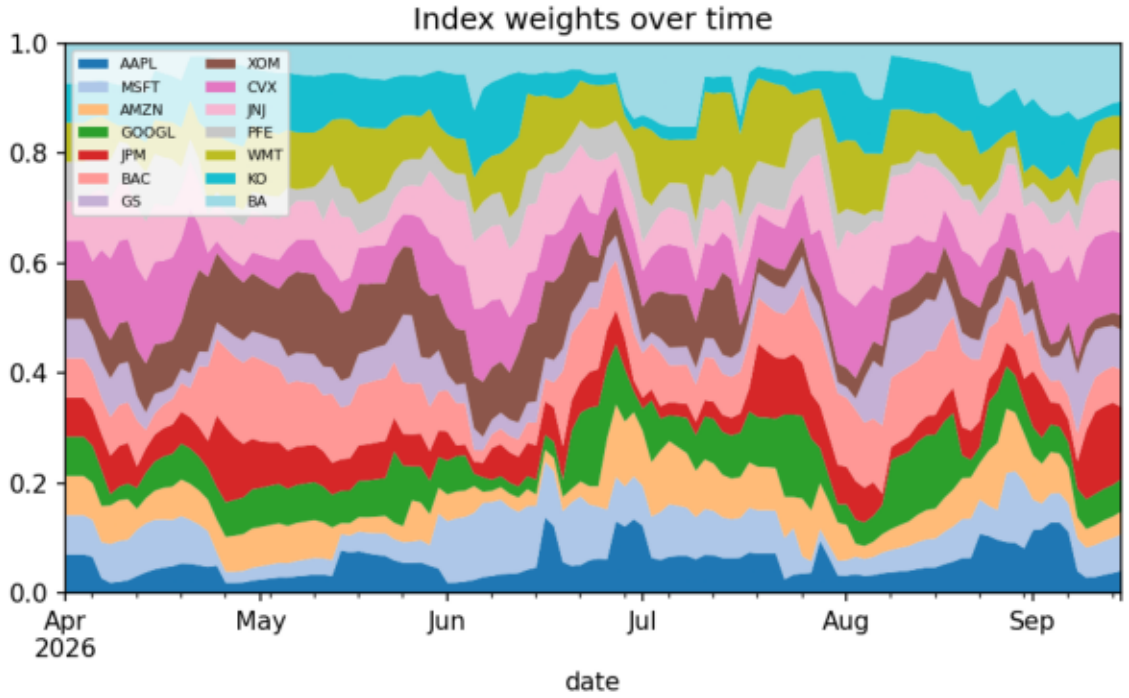
Module A: weight ~ exp(k * score) / N, bounded 2-15%, 10% turnover cap, 10 bps costs, no look-ahead
  Compared with equal-weight and buy & hold, plus a placebo test that shuffles signals across
    tickers.

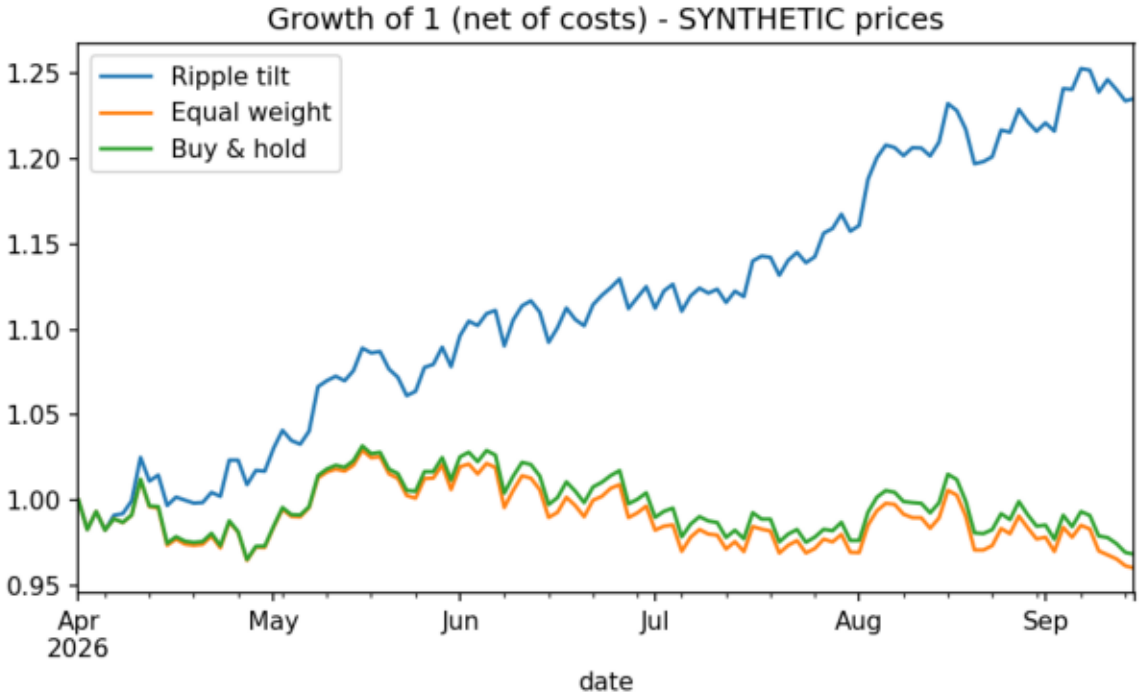
Module B: graph from event to sector/country, counterparty and asset; each instrument revalued on
    its own terms
  Loans: PD x LGD x EAD | Bonds: duration + convexity + credit loss | Derivatives: delta + CVA

Reliability: unit tests, a one-command pipeline and seeded, reproducible data

Key results - Module A (synthetic prices)

Sentiment-tilt: return 23.5%, Sharpe 3.12, max DD -2.9%, turnover 5.9%/day
Equal-weight: -4.0% Buy & hold: -3.2% Placebo percentile: 100%
CAVEAT: prices are synthetic and shocks follow events by construction - this validates the machinery, not alpha.
Re-run with real prices (fetch_real_data.py) for a genuine test.

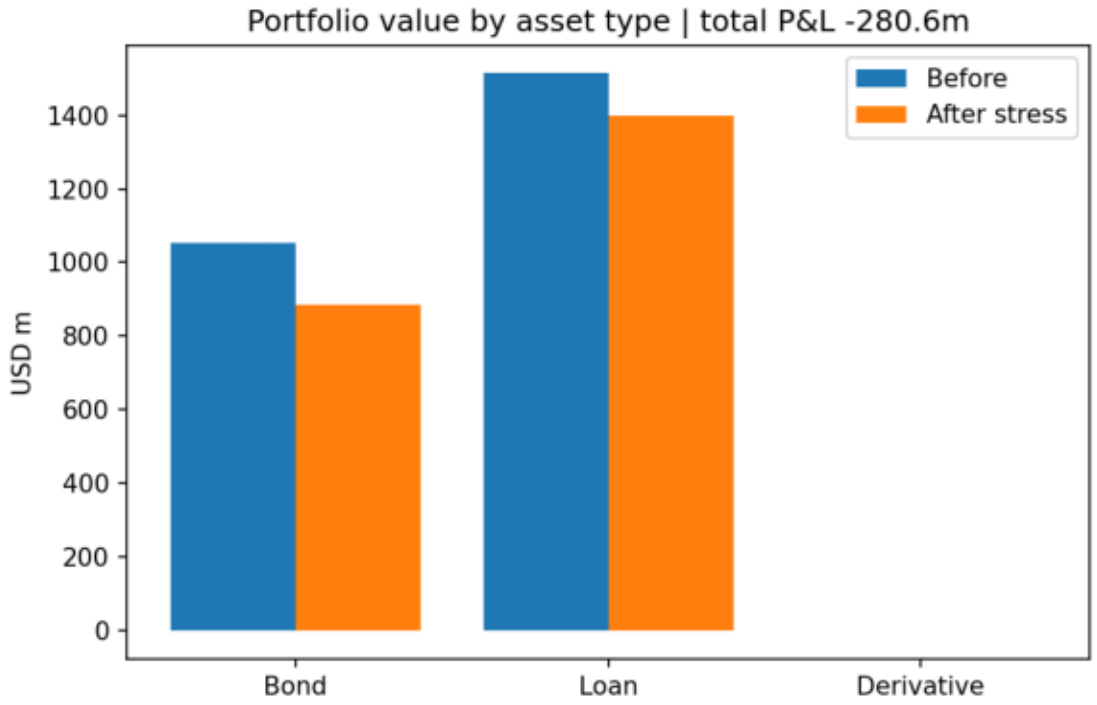


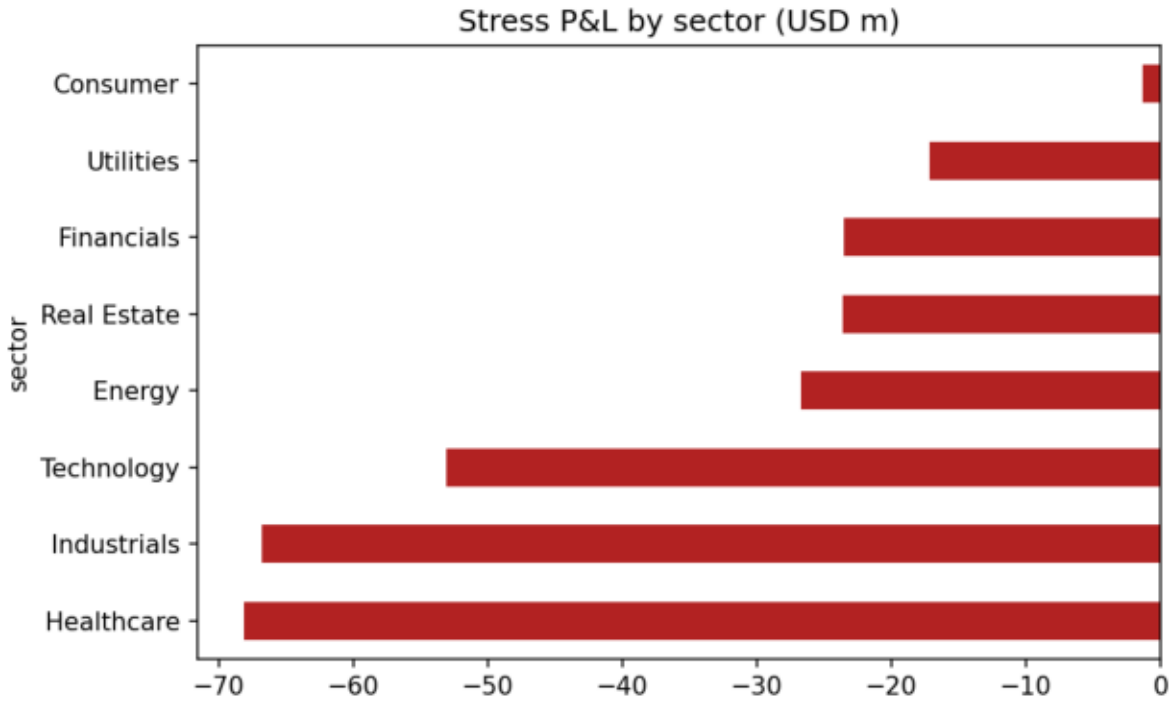


Key results - Module B

Auto-triggered scenario: portfolio 2,566m -> 2,285m (-10.9%). Naive flat shock would show only -114m and say nothing about WHERE the loss sits.

Replay: Banking Run -> -5.5%; Geopolitical Shock -> -5.2% | 1048 signals, 2 sources





Domain impact, limitations & next steps

Why it matters: risk teams receive auditable, machine-readable early warnings and an impact rating
based on observed market reaction;
stress tests follow actual exposure routes (sector, country, counterparty) instead of flat shocks.

Limitations: the demo uses synthetic data; the lexicon is coarse (FinBERT advised in production);
stress parameters are illustrative;
no intraday data; spill-overs are single-hop.

Next steps: live GDELT/NewsAPI streams, a spill-over matrix calibrated on past crises, per-
counterparty ratings,
Kafka ingestion, plus model monitoring and drift alerts.